

Fidelity Validation Against kinship2

Why this article exists

The [kinship2](#) R package is the field’s standard reference implementation for pedigree kinship computation and pedigree drawing (Sinnwell, Therneau & Schaid 2014). Its own supplementary material (Sinnwell, Therneau & Schaid, “The kinship2 R Package for Pedigree Data: Supplementary Material”) works a small, fully-specified 10-subject example pedigree through kinship2’s own kinship matrix, X-chromosome kinship matrix, and pedigree-trimming (“shrink”) functions.

The ratified [kinship2 supplement full-reproduction plan](#) closed 3 tracks against that supplement:

- **Track A** – `kinship()` gained `chrtype = c("autosome", "x")` and `sex` arguments, reproducing the supplement’s X-chromosome kinship matrix (Table S2).
- **Track B** – new `shrinkPedigree()`, a `kinship2::pedigree.shrink()` equivalent over this package’s own `id/sire/dam` data-frame pedigree representation.
- **Track C** – `makePedigreeMatingLayout()`’s consanguineous-mating visual marker (a distinct color/width on a blood-related couple’s mate-line edges) now propagates correctly onto `edgeStyle = "rectilinear"`’s dogleg-rerouted projection edges, not just the direct-style case.

Each track’s own implementing session (Track C: S563, Track A: S564, Track B: S565) verified its own fixtures against a live, installed `kinship2` 1.9.6.2 and recorded the results as hard-coded expected values in `tests/testthat/test_kinship.R`, `test_shrinkPedigree.R`, and `test_makePedigreeMatingLayout.R`. This article is the recorded, side-by-side evidence for a reader who was not in those sessions: for each track, the exact same fixture is run through **both** packages, live, and the numeric and graphic output is shown together, not just asserted equal in a test file. It is the same “validate before expose” discipline as [fg-se-validation.qmd](#), applied to a reference *package* instead of a reference *paper’s worked numbers*.

kinship2 is not a dependency of nprcgenomekeeper. Every comparison below was generated once, offline, by [data-raw/kinship2FidelityValidation.R](#) (`kinship2` installed locally, used interactively – matching the same evidence standard and the same “no new Suggests

dependency” choice the 3 tracks’ own implementing sessions already made) and the resulting numbers/images are embedded below, exactly as [fg-se-validation.qmd](#) embeds its own offline validation study’s results rather than recomputing them at render time.

```
# Reproduce (build-ignored; not run on render). Requires kinship2 installed
# locally (install.packages("kinship2")) and Chrome (chromote, an indirect
# Suggests dependency via shinytest2, for the nprcgenomekeeper diagram
# screenshots):
# Rscript data-raw/kinship2FidelityValidation.R
# Writes PNGs to vignettes/articles/kinship2-fidelity-validation-img/ and
# prints the numeric summary below to the console.
```

Track A – X-chromosome kinship (Table S2)

Fixture

The kinship2 supplement’s own Figure S1 subset (reconstructed from Table S1’s printed kinship values, per [KINSHIP2_SUPPLEMENT_REPRODUCIBILITY_AUDIT_2026-08-13.md](#)): a 10-subject pedigree in which subjects 8 and 9 are full siblings *declared* as a monozygotic (MZ) twin pair, and subject 10 is a child of twin 8 – the load-bearing case for confirming a twin correction propagates to a non-twin descendant, not just the declared pair.

Numeric fidelity

Both packages compute the full 10x10 autosomal matrix and the full 10x10 X-linked matrix on the identical fixture (nprcgenomekeeper’s own MZ-twin correction supplies the same `relation/twinRelations` declaration to each package in its own idiom):

Comparison	max nprcgenomekeeper – kinship2	Identical?
Autosomal kinship matrix (100 cells)	0	Yes
X-linked kinship matrix (100 cells)	0	Yes

A few named cells, reproducing the supplement’s own Table S2 (self-kinship differs by sex on the X chromosome – a male’s self-kinship is 1.0, not 0.5, since he carries a single X copy):

Pair	Relationship	nprcgenomekeeper	kinship2
(1, 1)	male self	1.0000	1.0000
(2, 2)	female self	0.5000	0.5000
(1, 3)	father-son	0.0000	0.0000

Pair	Relationship	nprcgenekeepr	kinship2
(1, 4)	father-daughter	0.5000	0.5000
(8, 9)	declared MZ twins, X-linked	1.0000	1.0000
(9, 10)	twin correction propagated to a child	0.5625	0.5625

Graphic fidelity

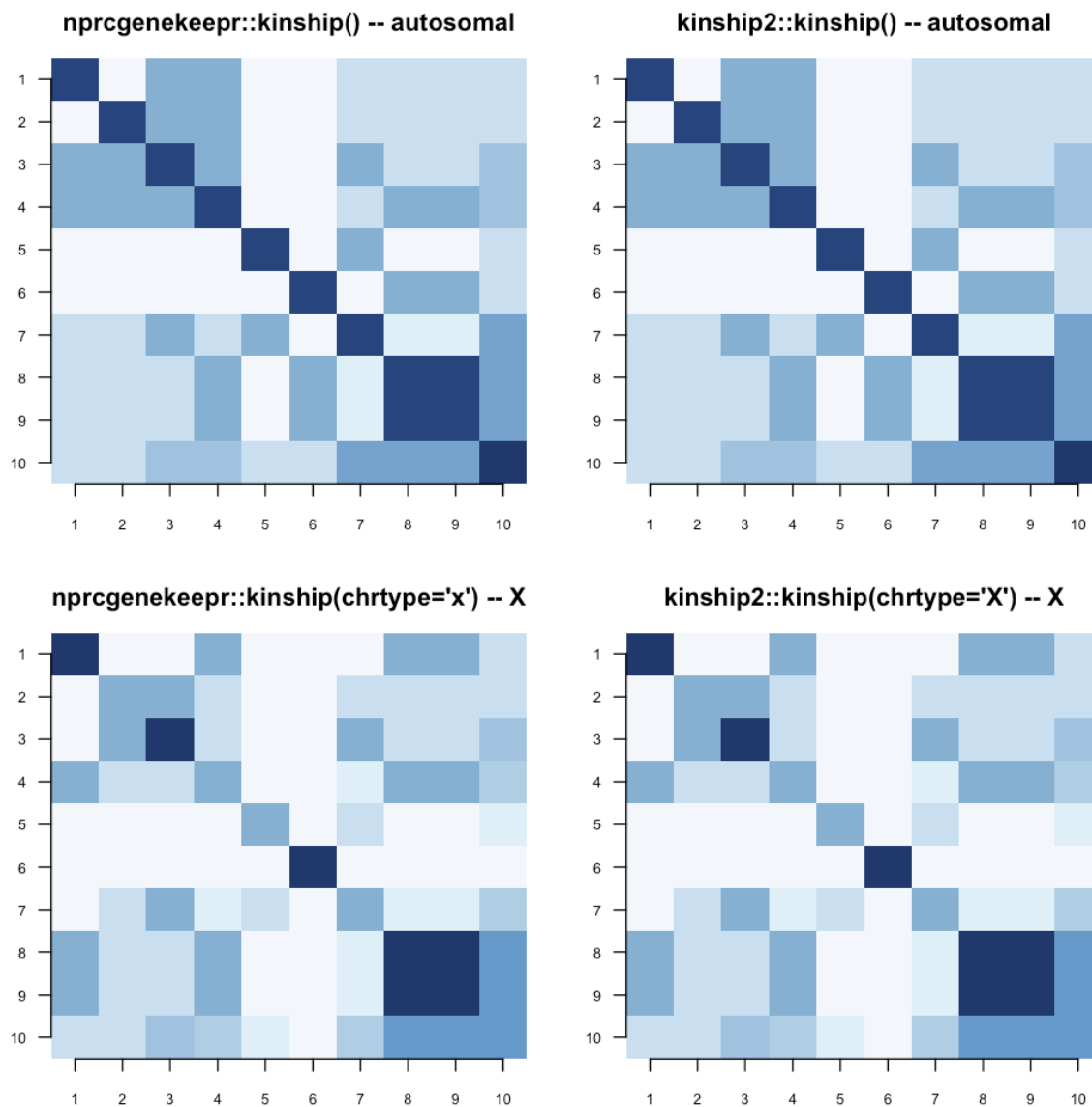


Figure 1: Kinship matrix heatmaps for the 10-subject fixture: `nprcgenekeeper`'s autosomal matrix (top left) against `kinship2`'s autosomal matrix (top right), and `nprcgenekeeper`'s X-linked matrix (bottom left) against `kinship2`'s X-linked matrix (bottom right). All 4 panels use the same color scale; the two autosomal panels are pixel-for-pixel identical, as are the two X-linked panels.

Track B – shrinkPedigree() vs. pedigree.shrink()

Fixture

The composite 16-subject fixture from `test_shrinkPedigree.R`, constructed to exercise every removal phase of kinship2’s own algorithm in one pedigree: an unavailable (ungenotyped) terminal leaf, an unavailable founder couple with a single child, a childless “stray marry-in” founder, a genotyped but non-informative individual, and a priority-ordered affected-status reduction down to `maxBits = 1`.

Numeric fidelity

Comparison	nprgenekeepr	kinship2	Match?
Surviving subject set (8 of 16)	C4, C4a, G3, L3, M1, P1, P2, P6	C4, C4a, G3, L3, M1, P1, P2, P6	Yes
bitSize trajectory	11 → 7 → 5 → 3 → 1	11 → 7 → 5 → 3 → 1	Yes

Graphic fidelity

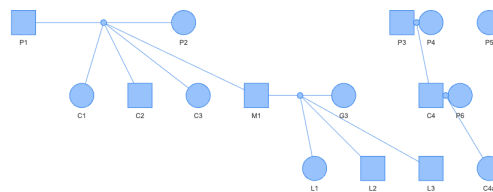
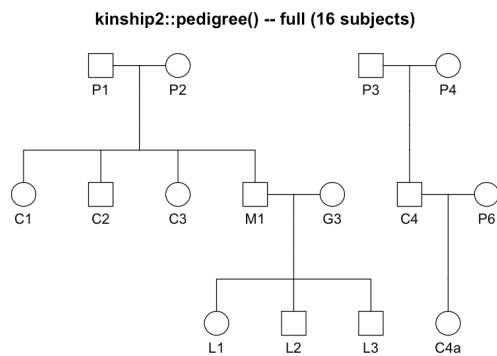
The same fixture, before and after shrinking, rendered by both packages. kinship2’s own `plot.pedigree()` did not plot subject P5 (an isolated, mate-less, child-less founder) – expected behavior for a disconnected singleton, not an error.

The two shrunk diagrams show the **same 8 surviving subjects in the same 2 family groups** – {P1, P2, M1, G3, L3} and {C4, P6, C4a} – confirming the numeric surviving-set match above is also structurally faithful, not just a matching id list.

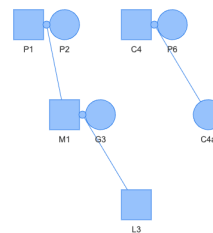
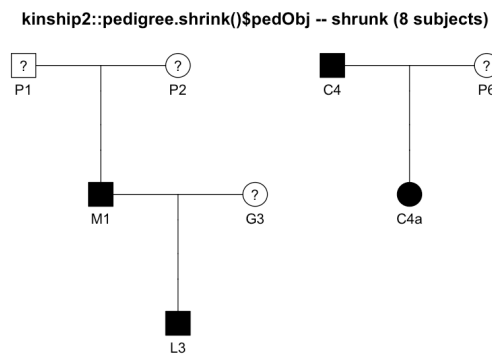
Track C – consanguineous-marker propagation

Fixture

The 9-subject dogleg fixture from `test_makePedigreeMatingLayout.R` (S563): A and Y are full siblings (children of P1 x P2) who mate with each other – the consanguineous union under test – while A also anchors an unrelated union with founder X, and Y also anchors an unrelated union with founder W. A’s own display generation is pulled away from the consanguineous union’s generation by the A x X union, forcing exactly one `edgeStyle = "rectilinear"` “dogleg” reroute on A’s side only.



- (a) kinship2's own `plot.pedigree()` on the full 16-subject fixture. (a) `makePedigreeMatingLayout()` + `visNetwork` on the same full 16-subject fixture.



- (a) kinship2's `pedigree.shrink()$pedObj` shrunk to 8 subjects. Unavailable ('?') individuals still shown, per kinship2's own convention. (a) `shrinkPedigree()`'s own surviving pedigree, same 8 subjects, rendered the same way.

Graphic fidelity

kinship2 draws a consanguineous mating as a **doubled connecting line** between the two mates (visible directly between A and the left-hand Y) and – independently of anything to do with consanguinity – duplicates any individual who appears in more than one union (Y appears twice, joined by a dashed connector). `makePedigreeMatingLayout()` uses a **distinct color and width** on the consanguineous union’s own edges and *also* independently duplicates a multi-union individual with a dashed connector – the two packages converge on the same duplicate-node convention without having copied it from one another.

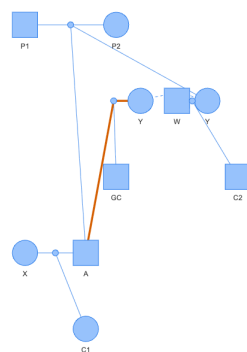
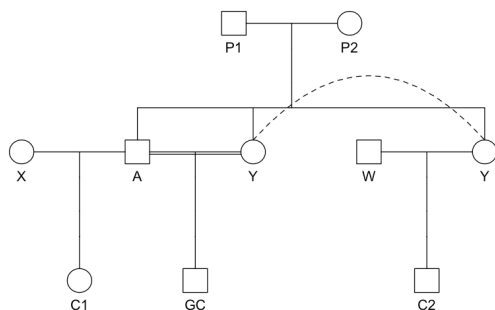
Edge style	Marked (vermillion) edges	Expected
<code>direct</code>	2	2 (one edge per mate, no dogleg)
<code>rectilinear</code>	3	3 (A’s 1 edge splits into 2 dogleg segments; Y’s 1 edge is unaffected)

Both packages flag the *same* union as consanguineous, using their own independent visual conventions – a thickened doubled line in kinship2, a distinct color and width in nprcgenekeeper – and nprcgenekeeper’s marker survives the more complex rectilinear dogleg reroute exactly as it does the simple direct case.

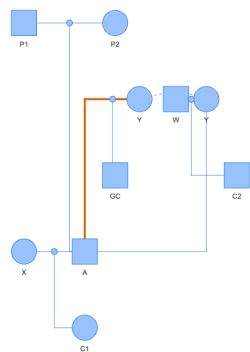
Caveats carried forward

- **kinship2 is not a package dependency.** Nothing above runs at `quarto render` time or in R `CMD check` – the numbers and images are the frozen output of one offline, interactively-run script, matching this package’s established precedent of never calling `kinship2::` live from committed test or documentation code.
- **The full 17-subject fam1 pedigree from the kinship2 supplement’s main worked example is not reconstructable from this repository’s materials** (its source figure lives in the kinship2 *application note*, not the supplement PDF this repository ships). Track A’s fixture is the fully-specified 10-subject Figure S1 *subset*, per the audit’s own scope caveat – not a limitation introduced by this article.
- **Tracks B and C have no PDF-printed worked example at all** – the supplement names only *which* subjects a shrink trims, never their relationships, and says nothing about visual conventions. Both tracks’ ground truth is a live, installed `kinship2::pedigree.shrink()` / `plot.pedigree()` run on a fixture purpose-built to exercise the relevant algorithm, not a supplement-sourced value.
- **kinship2::pedigree()’s sex-role validation is stricter than nprcgenekeeper’s own sire/dam columns.** Track C’s own committed test fixture lists one individual (Y) as a *sire* in one row despite her declared sex being female – valid input to

kinship2::pedigree() -- A x Y consanguineous (doubled line)



- (a) `kinship2`'s own `plot.pedigree()`. The doubled line directly between A and the near Y node is `kinship2`'s consanguinity marker; the dashed arc connects Y's two appearances (one per union she anchors).
- (a) `makePedigreeMatingLayout(edgeStyle = "direct")` – the A-Y union's 2 mate edges render in vermilion (`#D55E00`) at width 4.



- (a) `makePedigreeMatingLayout(edgeStyle = "rectilinear")` – the same marker now propagates onto A's dogleg-rerouted projection edges (this session's own fix), not just Y's non-doglegged side.

`makePedigreeMatingLayout()`, which does not enforce sex/column-role consistency, but rejected by `kinship2::pedigree()`, which does. The validation script swaps that one row's 2 column values (same 2 parents, same family structure) only for the kinship2-side object; every `nprcgenomekeep` call in this article uses the fixture exactly as committed.

Verdict

PASS, all 3 tracks. Track A's autosomal and X-linked kinship matrices are bit-for-bit identical to kinship2's own output across every one of 200 compared cells (100 autosomal + 100 X-linked), including the MZ-twin correction and its propagation to a descendant. Track B's `shrinkPedigree()` reproduces kinship2's exact surviving subject set and exact `bitSize` trajectory, and the shrunk pedigrees are visually the same 2 family groups. Track C's consanguineous-mating marker flags the same union kinship2 flags, under both edge styles, using an independently-converged duplicate-node convention for the same underlying multi-union case kinship2 also duplicates. All 3 tracks are cleared as faithful reproductions of the kinship2 supplement's own results, not just internally self-consistent implementations.

References

Sinnwell, J.P., Therneau, T.M., Schaid, D.J. (2014) "The kinship2 R package for pedigree data." *Human Heredity* 78(2):91-93.

Sinnwell, J.P., Therneau, T.M., Schaid, D.J. "The kinship2 R Package for Pedigree Data: Supplementary Material." Mayo Clinic (PMC manuscript NIHMS593658); shipped in this repository at `inst/extdata/reference/NIHMS593658-supplement-supplement_1.pdf`.

See also `kinship()`, `shrinkPedigree()`, `makePedigreeMatingLayout()`, and [kinship2-supplement-full-rep](#)